

What is Hyperthyroidism?

- A condition where the thyroid gland produces excess thyroid hormones (T3 and T4).
- Leads to a hypermetabolic state affecting multiple body systems.
- Affects about 1-2% of the population, more common in women (5-10x).

Thyroid Physiology: The Hormonal Feedback Loop

- Hypothalamus releases TRH → stimulates pituitary to release TSH.
- TSH stimulates thyroid gland to produce T3 (active) and T4 hormones.
- Negative feedback: high T3/T4 suppress TRH and TSH release.
- Thyroid hormones regulate metabolism, growth, heart rate, and temperature.

Major Causes of Hyperthyroidism

Graves' Disease (60-80% of cases): Autoimmune stimulation of TSH receptors by antibodies.

Toxic Multinodular Goiter: Multiple thyroid nodules autonomously producing hormones.

Toxic Adenoma: Single overactive thyroid nodule.

Other causes: Thyroiditis, iodine excess, medication-induced (e.g., amiodarone).

Clinical Symptoms and Signs

Common symptoms: weight loss despite appetite, heat intolerance, sweating, palpitations, anxiety, tremors.

Physical signs: diffuse goiter, tachycardia, exophthalmos (bulging eyes in Graves'), pretibial myxedema.

Variable presentation: some patients may have subtle or no symptoms initially.

Diagnostic Approach

Blood tests: suppressed TSH with elevated free T4 and/or T3 confirms hyperthyroidism.

Antibody testing: TSH receptor antibodies confirm Graves' disease.

Imaging: thyroid ultrasound and radioactive iodine uptake scan to identify nodules or inflammation.

Treatment Options

Antithyroid medications (methimazole, propylthiouracil) to block hormone synthesis.

Radioactive iodine therapy to ablate overactive thyroid tissue (most common in the US).

Surgery (thyroidectomy) for large goiters, nodules, or when other treatments fail.

Beta-blockers to control symptoms like rapid heart rate and tremors.

Special Considerations

Thyroid storm: life-threatening exacerbation with fever, tachycardia, delirium; requires emergency care.

Pregnancy: careful management needed; some antithyroid drugs are preferred.

Monitoring for hypothyroidism after treatment is essential.

Case Highlight: Graves' Disease

Autoimmune disorder causing diffuse thyroid enlargement and eye disease.

TSH receptor antibodies mimic TSH → excessive hormone production.

Up to 30% develop Graves' ophthalmopathy with eye bulging and discomfort.

Requires multidisciplinary care including endocrinology and ophthalmology.

Summary & Takeaway

Hyperthyroidism is a common, treatable endocrine disorder with systemic effects.

Early diagnosis and tailored treatment prevent complications.

Understanding causes and symptoms helps improve patient outcomes.

Stay vigilant for thyroid storm and special populations like pregnant women.